# *Control and Stability Analysis of DC Micro-grids with Constant Power Loads and Source Disturbances*

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***Abstract*- Grid connected DC Microgrid (DCMG) with wind turbine and solar PV distributed generations (DG), battery storage system and constant power loads (CPL) which are integrated via power electronic converter is presented in this paper. In DCMG interfaced with multiple parallel converters the stability of dc bus can be deteriorated by DG output power intermittency and microsources converter output impedance and load converter input impedance mismatches. Thus to solve these problems, this paper presents an autonomous control using decentralized control scheme based on dc bus voltage as means of communication is proposed in this paper. Droop control strategy for power sharing among the different sources and to stabilize bus voltage is employed. The stability analysis of the system is performed using Lyapunov stability criteria to determine the qualitative behavior of the system based on the simplified equivalent source transmission line. The control method verified via MATLAB/SIMULILK simulation.**

***Keywords- CPL, Droop control, DC Microgrid, Stability Analysis***

## I. INTRODUCTION

Microgrid is a small power network system for building applications such as in residential, academia and industrial sectors. They can be categorized as DC, AC or DC/AC hybrid microgrids. DC micro grids (DCMG) are based on direct current technologies and consisting of distributed energy resources and loads operating in controllable and synchronized way either in grid-connected or islanded manner [1]. Due to the advancement in the semiconductor electronic switches and power electronic converters nowadays, microgrids are becoming popular topology with the integration of DG, energy storage system (ESS) and dc loads [2], [3]. To date, DCMG is getting more preference than its ac system because of numerous advantages such as direct connection of native dc loads with renewables natively generate low voltage dc power, easier control system compared to ac counterparts, higher reliability and efficiency [4] – [7]. For a DCMG with various sources an independent and coordinated control for stable and proper operation is required. Thus, for the purpose of assuring stability operation of DCMG, there are three control techniques are reported in literature in accordance with communication prospective [7]; centralized [8], [6], distributed [7] and decentralized [9]- [10] methods. In case of centralized control, the system is aware of each node and each source is controlled centrally from one point via a communication link. Hence, this mechanism degrades the reliability of the DCMG since it is depend on communication link for proper control operation of the system. [11] - [13]. This control method degrades the reliability of the system since it employs central control technique which may lead to poor system operation in case of any key communication link failures. In distributed control method the system is not controlled centrally, but the local control systems communicate among themselves via some dedicated digital communication links (DCLs). In this control strategy, the system can maintain full functionalities in case of certain communication links failure. However the system is characterized by limitations such as complex analytical performance analysis, communication time delays and measurement errors [7].

Thus, to solve the problems mentioned in the central and distributed control methods, a decentralized control approach is implemented in this paper. Decentralized operation is convoyed by mode transition based operation so that the distributed sources in DCMG can cope up with the bus voltage and power flow control independently by using local variables of the interfacing converter [14]. In DCMG, loads can be classified as resistive and CPL. CPL is typically maintained via point of load (POL) converter with a sufficiently large bandwidth. It should be noted that load power is equal to the POL converter input power regardless of the source voltage deviations [6], [15]. The influence of CPL on the system stability is brought by its negative input resistance in the small-signal analysis. This negative incremental impedance causes the system badly damped and can cause instable poles in the frequency domain and worsen the system stability [8]. In order to solve instability problems caused by nonlinear CPL an admittance or impedance based design is proposed in [16]. In [10] stability analysis is based on small signal and damping enhancement using frequency dependent virtual impedance for DCMG with presence of CPL is introduced. Similarly in [8] stability improvement using virtual impedance was proposed. Most of the DCMG stability analysis reported in the literature are linear and based on small signal analysis [8], [17].

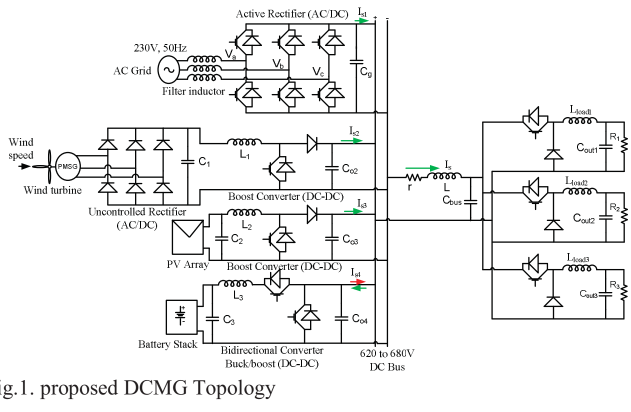

Fig.1. proposed DCMG Topology

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However, due to the intermittency characteristics of DGs and the negative input resistance behavior of CPLs DCMG with multiple sources supplying CPL is nonlinear system operating with different voltage mode. In this paper, a decentralized control structure based dc bus signal is proposed for coordinated and stable operation of DCMG system. This approach ensures fully self-governing control action of each converter terminals without communication. In order to solve the stability problems, a nonlinear stability analysis is presented in this paper to predict the system's global stability conditions. The proposed topology of the DCMG is shown in Fig.1. The system topology consists of wind turbine, solar PV, battery energy storage, grid and load subsystems.

The remaining part of the work in this paper is outlined as follows: section II explains the proposed system control methodology. In section 3 describes the power flow management strategy of the system and in section 4 the dynamics of the DCMG is described. A principle of operation of the system detail is given in section 5 and section 6 describes stability analysis of the system. Section 7 explains simulation outcome and section 8 summarize the work.

## II. PROPOSED SYSTEM CONTROL METHODOLOGY

### *A. Source subsystem control*

Two control techniques are employed for microsource converters; namely: voltage control and constant power control approaches. In the voltage control mode, the droop constant imitates an impedance comportment which reduces the converter output voltage with the raise of the source current. Fig.2 (a) illustrates the block diagram of droop controlled microsources. The voltage and current droop coefficients for each converters is calculated as the following [18], [19]:

$$V_{kr} = V_k - r_{dn} i_o \tag{1}$$

Where $r_{dn}$, is droop constant for n converter (n = 1, 2,…, n$^{th}$, n refers to no of converter ), $V_k$ is the converter terminal voltage at no load and $i_o$ is converter output rated current. Let $\Delta V$is the permitted voltage variation limit, then $r_{dn}$ can be found by:

$$\mathrm{r_{dn}} = \frac{\Delta V}{\mathrm{i_{max}}} \tag{2}$$

Where $i_{max}$ is the maximum rated current. The load allocation among interfacing converter is given by:

$$\Delta \mathrm{i_o} = V_{kr}/\mathrm{r_{dn}} \tag{3}$$

The constant power control can also be categorized into two; namely constant power supply source (CPSS) and constant power supply load (CPSL). The sources which are injecting constant power into the dc grid considered as CPSS. Examples: solar photovoltaic and wind-turbine in constant power control mode, battery in discharging mode and grid in rectification mode. The CPSL are units that absorbing constant power from dc bus; (e.g., battery in charging mode and grid in inverting mode). The control technique for constant power control is depicted in Fig.2b. Mathematically, $I_{CPSS}$ and $I_{CPSL}$ can be calculated as in (4) [18].

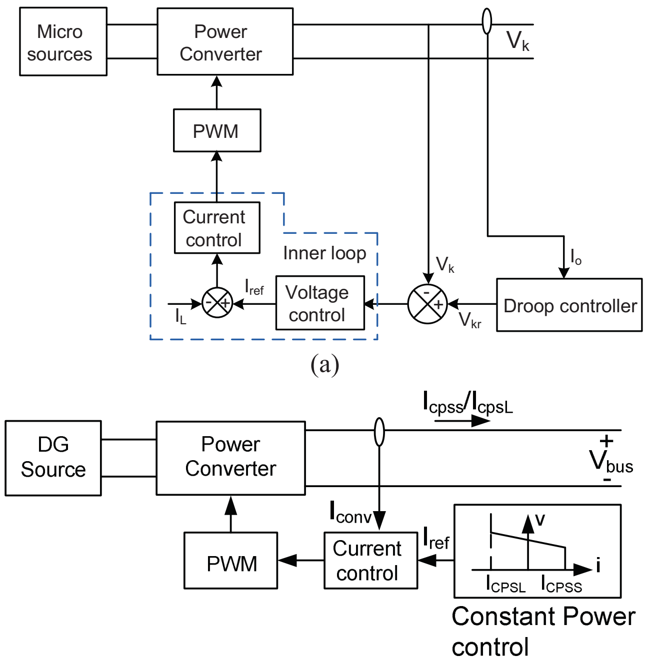


Fig.2: (a) Voltage Control Mode, (b) Constant Power Control Mode

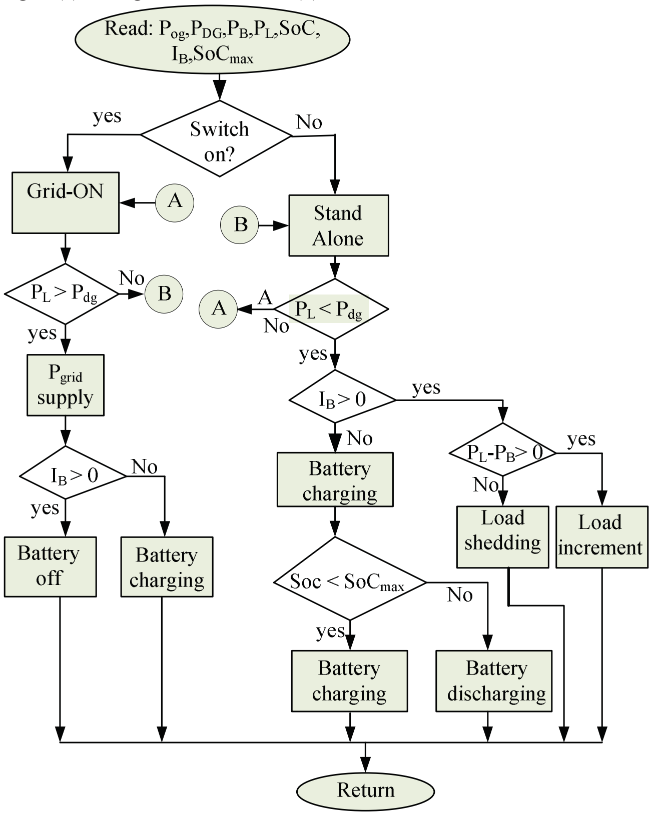


Fig.3: Power Flow Control Algorithm [20]

$$I_{cpss} = \frac{P_{cpss}}{V_{bus}}\,;\, I_{cpsL} = \frac{P_{cpsL}}{V_{bus}} \tag{4}$$

### *B. Constant Power Load modeling*

In DCMG every load requires a particular voltage value for suitable working. This in turn makes the system to have point of load converters which can be tightly well-ordered to maintain output voltage constant. It is also assumed that the output power of point of load converter is equal to the input power. The average current mode control method is employed for POL converter control as shown in Fig.4. The load converters perform as a CPL, which can be defined by a voltage controlled current source and mathematically given by [6]:

$$I_{CPL} = \frac{P_{CPL}}{V_{bus}} \tag{5}$$

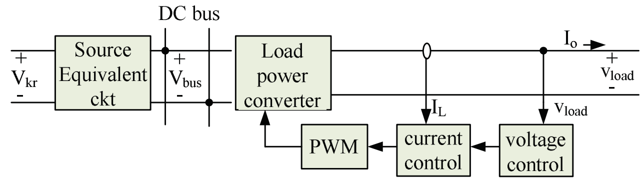


Fig.4: Load Converter Control Block Diagram

## III. THE PROPOSED POWER MANAGEMENT TECHNIQUES

A control algorithm for power balancing of dc micro grid under varying power generation by DGs output and load change is anticipated in this research is illustrated in Fig.3. In the presented DCMG the power from grid and battery is used only when generation by wind turbine and solar PV is unable to satisfy the desired load requirement. The power flow control algorithm is described in the following steps.

Step1: Check power generation by wind turbine and solar PV. If $P_L < P_{DG}$

$$\begin{cases} \text{Yes; go to step 2} \\ \text{No; go to step 5} \end{cases}$$

Step 2: Check battery current. If $I_{bat} > 0$

$$\begin{Bmatrix} \text{Yes; go to step 3} \\ \text{No; battery charging} \\ \text{and go to step 4} \end{Bmatrix}$$

Step 3: Check battery power and total load demand

If $P_L - P_B > 0$

$$\begin{Bmatrix} \text{Yes; Increase load} \\ \text{No; decrease load} \end{Bmatrix}$$

Step 4: Check battery state of charge. If $SOC - SOC_{max} < 0$

$$\begin{Bmatrix} \text{Yes; Battery charging} \\ \text{No; Battery discharging} \end{Bmatrix}$$

Step 5: Check total DG power and power demand.

If $P_L > P_{DG}$

$$\begin{Bmatrix} \text{Yes; grid supply and go to step 6} \\ \text{No; Islanding and go to step 2} \end{Bmatrix}$$

Step 6: Check battery current status. If $I_{bat} > 0$

$$\begin{cases} \text{Yes; Battery disconnected} \\ \text{No; Battery is charging} \end{cases}$$

Step 7: If generation by DGs (wind and solar) and demands are changing, then repeat all the steps (1-6).

## IV. DYNAMIC EQUATION OF THE PROPOSED DC MICRO GRID

Under droop control every converters are designed as a Thevenin equivalent voltage source $V_{kr}$ in series with virtual droop constant $r_{dn}$ and transmission line impedance (resistance $r_{ci}$ in series with inductance $L_{ki}$). The reduced model circuit of the proposed DCMG is depicted in Fig.5. The droop impedance and transmission line resistances are in series and can be evaluated mathematically as given in (6) [21].

$$r_{ki} = r_{dn} + r_{ci} \tag{6}$$

For stable equilibrium points to exist at large transients, it should be noted that the source output resistance must be designed to be larger than the simplified line resistance. As a result the virtual droop resistance desired to be much greater than the line resistance ($r_{dn} \gg r_{ci}$), indicating that:

$$r_{ki} \approx r_{dn} \tag{7}$$

Assuming the same reference voltage $V_{kr}$ for all interfacing source converters coupled in parallel, the following condition should be met:

$$V_{kr} = V_{k2} = V_{k3} = V_{k4} = V_k \tag{8}$$

With the condition

$$\frac{r_{k1}}{L_{k1}} \approx \frac{r_{k2}}{L_{k2}} \approx \cdots \approx \frac{r_{kn}}{L_{kn}} \tag{9}$$

The dc-link capacitances of the POL converters linked to the dc bus are represented by lamped $C_k$ (the sum of all capacitance in parallel). Thus, the dynamic differential equation of the DCMG can be described by [18], [21]:

$$\begin{aligned} \frac{di_{k1}}{dt} &= \frac{(V_{kr} - v_{bus})}{L_{k1}} - \frac{i_k(r_{k1} + r_{d1})}{L_{k1}}, \\ \frac{di_{k2}}{dt} &= \frac{(V_{kr} - v_{bus})}{L_{k2}} - \frac{i_{k2}(r_{k2} + r_{d2})}{L_{k2}}, \\ \frac{di_{kn}}{dt} &= \frac{(V_{kr} - v_{bus})}{L_{kn}} - \frac{i_k(r_{kn} + r_{dn})}{L_{kn}} \end{aligned} \tag{10}$$

The total input current is supplied by the sources is found by:

$$i_k = i_{k1} + i_{k2} + i_{k3} \ldots + i_{kn} \tag{11}$$

Thus the total supply current of the CPLs $i_k$ mathematically defined with $n$ differential equation is given by:

$$\frac{di_k}{dt} = \frac{d}{dt}\sum_{i=1}^{n} i_{ki} \approx \left(\sum_{i=1}^{n}\frac{1}{L_{ki}}\right)(V_{kr} - v_{bus}) - \sum_{i=1}^{n}\frac{r_{ki}}{L_{ki}} i_{ki} \tag{12}$$

The ratio of $\frac{r_{ki}}{L_{ki}}$ in (12) from all situations satisfying the condition $\frac{r_{k1}}{L_{k1}} \approx \frac{r_{k2}}{L_{k2}} \approx \cdots \approx \frac{r_{kn}}{L_{kn}}$, can be estimated by the ratio $R_\alpha$ and $L_\alpha$ where $R_\alpha$ and $L_\alpha$ are the average mean of $r_{ki}$ and $L_{ki}$, given by:

$$R_\alpha = \frac{\sum_{i=1}^{n} r_{ki}}{n}\ ;\ L_\alpha = \frac{\sum_{i=1}^{n} L_{ki}}{n} \tag{13}$$

Multiplying both sides of (12) by (14), equation (15) is obtained.

$$L_k = \frac{1}{\sum_{i=1}^{n}\frac{1}{L_{ki}}} \tag{14}$$

$$L_k \frac{di_k}{dt} \approx (V_{kr} - v_{bus}) - R_\alpha \frac{L_k}{L_\alpha} i_k \tag{15}$$

Thus the single equivalent circuit is approximated analytically described by (16) & (17).

$$L_k \frac{di_k}{dt} \approx V_{kr} - v_{bus} - r_k i_k\ ,\ r_k = R_\alpha \frac{L_k}{L_\alpha}$$

$$L_k \frac{di_k}{dt} = V_{kr} - i_k r_k - v_{bus} \tag{16}$$

$$\frac{dv_{bus}}{dt} = \frac{i_k}{C_k} - \frac{P_o}{C_k V_{bus}} \tag{17}$$

Where

$$\begin{cases} i_k = i_{k1} + i_{k2} + i_{k3} + i_{k4} \\ r_k = \frac{1}{\sum_{i=1}^{n}\frac{1}{R_{dn}}} \\ P_o = P_{CPL} + P_{CPSL} - P_{CPSS} \end{cases}$$

## V. OPERATION OF DC MICROGRID

In decentralized control method of DCMG there are different voltage levels which vary in some specified acceptable limit. For this paper, the nominal bus voltage $V_k$ is set at $650\ V$, $\delta_{h1}$ and $\delta_{h2}$ are designed at 2.5% and 5% higher than $V_k$ respectively, whereas; $\delta_{L1}$ and $\delta_{L2}$ set at 2.5% and 5% below$V_k$, respectively. In this paper, wind and solar terminal voltages are designed at $V_k + \delta_{h2}V_k$ and $V_k + \delta_{h1}V_k$ respectively. Whereas the battery storage and grid are designed at $V_k - \delta_{L1}V_k$ and $V_k - \delta_{L2}V_k$ and where

$\delta_{h2}$, $\delta_{h1}$, $\delta_{L2}$ and $\delta_{L1}$ are parameters which determines the voltage levels [18].

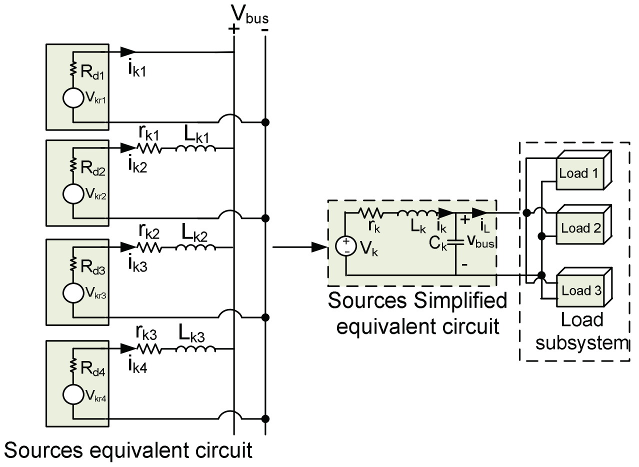


Fig.5: Simplified Equivalent Circuit of Proposed System

TABLE 1 MICROSOURCES DATA

| Source units | Power rating (kW) | Droop coefficient | Nominal voltage (V) |
|---|---|---|---|
| Wind | 7 | 1.3077 | 682 |
| PV | 14 | 0.6711 | 682 |
| G-VSC | 7 | 1.5089 | 650 |
| Battery | 5 | 1.5089 | 634 |

Accordingly, the acceptable functioning voltage limits for DC bus fall in the interval of $620\,V$ - $682\,V$, to enable for load transfer and voltage control via droop controller scheme. All the interfacing converters are well-ordered to switch on/off in the specified voltage margins at predetermined time. Thus, in this paper, three different voltage modes are designed (DG, G-VSC and battery storage voltage modes). Based on these three different modes, three POL converters are interfaced in parallel to the DC grid due to each load in DCMG works properly with a specific voltage level [19]. The Microsources design parameters used in this paper is shown in Table 1.

## VI. SYSTEM STABILITY ANALYSIS

The equilibrium points are achieved by making the equations given in (16) and (17) equals to zero and given by [22]:

$$V = \frac{V_k \pm \sqrt{V_k^2 - 4P_oR_k}}{2}, \; I_k = \frac{P_o}{V} \tag{18}$$

Thus to circumvent complex steady state points, the conditions $P_o < \frac{V_k^2}{4r_k}$ have to be satisfied [18], [22]. This constraint keeps the upper boundary on the source impedance as per the effective voltage $V_k$ and the quantity of constant power $P_o$ to be brought by the source [22]. In order to apply the Lyapunov approach in the proposed system, assume that the input voltage $V_k$ is unchanged and the steady state points move the coordinate to the origin [18]. Considering

$$x_1 = i_k - I_{ko},$$

$$x_2 = v_{bus} - V_{buso} \tag{19}$$

Then (16) & (17) is linearized and the Jacobian matrix A for the subsystems is achieved as in (19).

$$\begin{bmatrix} \dot{x_1} \\ \dot{x_2} \end{bmatrix} = \begin{bmatrix} -\frac{r_k}{L_k} & -\frac{1}{L_k} \\ \frac{1}{C_k} & \frac{P_o}{C_k(x_2+V_{bus})^2} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = A_i \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \tag{20}$$

Where $i = 1, 2, 3$.The system is said to be asymptotically stable near the steady state point $x_o$ if it is Lyapunov stable and there exist a positive definite matrix P and I, and which must satisfying the following condition [18]:

$A_i^T P + PA_i = -I$ and $||x|| \to \infty$, then the Lyapunov function

$$V(x) = x^T P x \to \infty \tag{21}$$

then system in (20) said to be globally asymptotic stable. If I is identity matrix then the matrix P can be found by:

$$P = \frac{-0.5*[I+\det(A_i)(A_iA_i^T)^{-1}]}{tr(A_i)} \tag{22}$$

Where $tr(A_i) = -\frac{r_k}{L_k} + \frac{P_o}{C_k(V_{co})^2}$

It can be understood that the subsystem is stable when the determinant of the matrix $P > 0$. In addition the following two conditions also must simultaneously satisfied:

$$tr(A_i) < 0 \text{ and } det(A_i) > 0 \tag{23}$$

Similarly, for the subsystem to be stable at the equilibrium point the following two conditions necessarily fulfilled:

$$P_L < \frac{V^2}{r_k}, \; C_k > \frac{L_k * P_o}{r_k * V^2} \tag{24}$$

From the above expressions of (24) it can be observed that larger value of $C_k$ and smaller of $P_L$ enable to achieve a stable equilibrium point in each subsystem. According to the aforementioned approaches, to predict system stability conditions simulation was performed at 2.24kW, 17kW, 9.24kW of load power with mode 1, 2, and 3 respectively and the values of $L_k, C_k$ and $r_k$ are chosen to be 51μH, 1.1mF and 0.1438Ω respectively.

## VII. SIMULATION RESULT AND DISCUSION

Simulation is performed in MATLAB/Simulink to validate the proposed control method. The simulation is done based on three different mode of operation in this work. These are DG mode (mode 1), grid interfaced mode (mode 2) and energy storage mode (mode 3). The system parameters used in the simulation is shown in Table 2 in the appendix. In Mode 1 and 3 the system is in islanding mode. In mode 1 during t = 0-3 sec, grid is disabled as illustrated in Fig.7: (a). The load demand is less as observed from Fig.7 (c). Due to light-load during this period, generation by DG is more than load demand. As result the excess power from the DG is supplied to charge the battery as shown in Fig.7 (d) & Fig.8 (b & c). In this mode DGs are extracting the maximum available energy with MPPT control method. As can be seen from Fig.6 the DC bus voltage is operational in the intervals 665V to 682 V. In grid-connected mode from t = 3–7sec, the DG generation is operating at full MPPT with increased output power from 7.24kW to 18kW (Fig.7b).

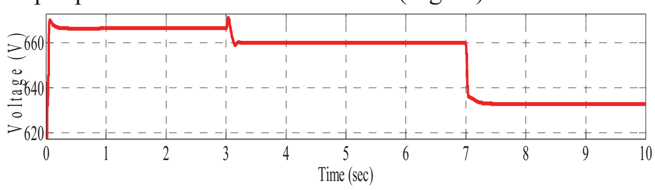


Fig.6: DC Bus Voltage

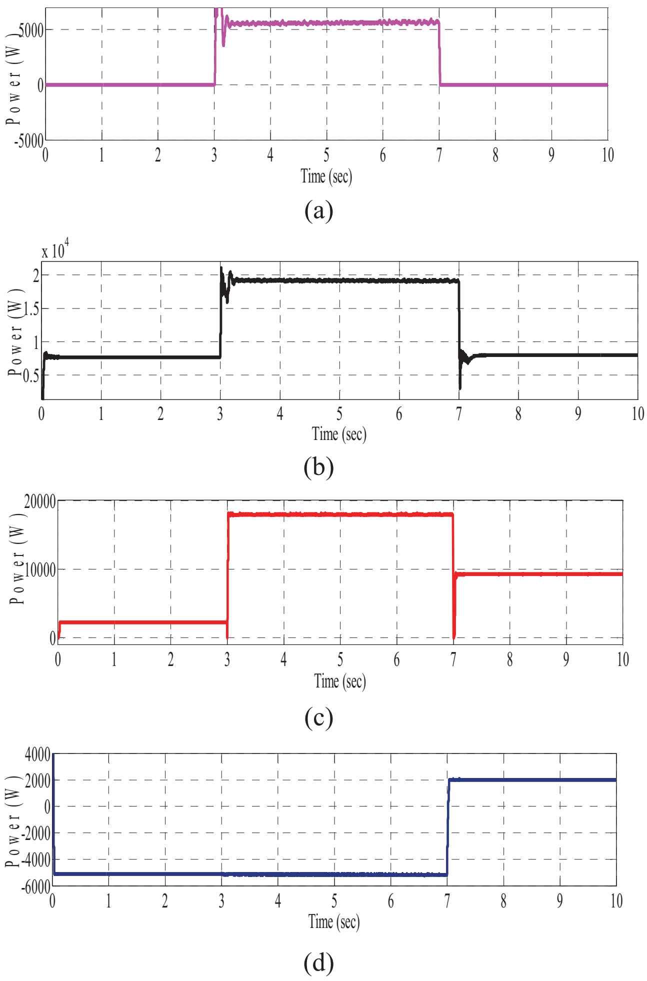


Fig.7: (a) Grid power, (b) Total DG Power, (c) Load Power, (d) Battery Power.

Similarly load demand also increased from 2.24kW to 17kW (Fig.7c). Battery is in charging mode and consuming 5kW (see Fig.7d). Since demand is greater than generations the DCMG is connected to grid to balance the mismatch and grid is supplying 5.2kW (see Fig.7 (a)) power to the load. During this period, dc-bus voltage operates in the intervals of 634V- 665V as shown in Fig.6. In Fig.9 (a) & (b) similarly during the time t= 3-7sec, solar and wind are working with 1000 (W/m^2) irradiance and 12(m/sec) wind-speed respectively.

In mode 3 when t = 7–10sec, grid is disconnected and generation also decreased from 18kW to 7.24kW as indicated on Fig.7 (b). In this mode, demand is larger than generation by wind turbine and solar PV (Fig.7 (b) & (c)). At this time, battery is shifted to discharging mode in order to balance the power difference between demand and generation (see Fig.8 (b) & (c)).

In Fig.6 it is shown that the operating dc-bus voltage is in the intervals of 620 V to 634 V. In Fig.7(d) shows that battery is supplying 2kW to fulfil the power deficient by the load. In all the three modes, the output of POL converter voltage is maintained constant at 220 V as shown in Fig. 8 (a).

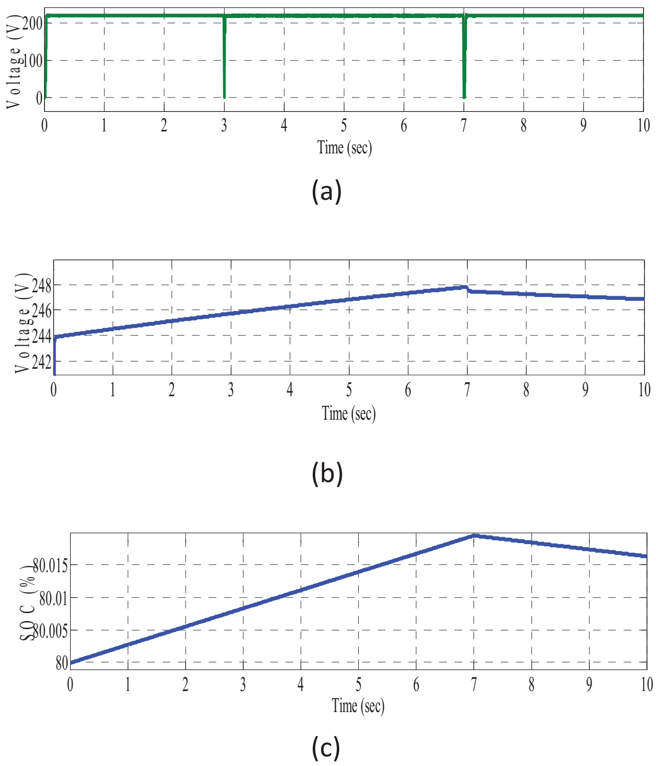


Fig.8: (a) Load Voltage (V), (b) Battery Voltage (V), (c) Battery SOC (%)

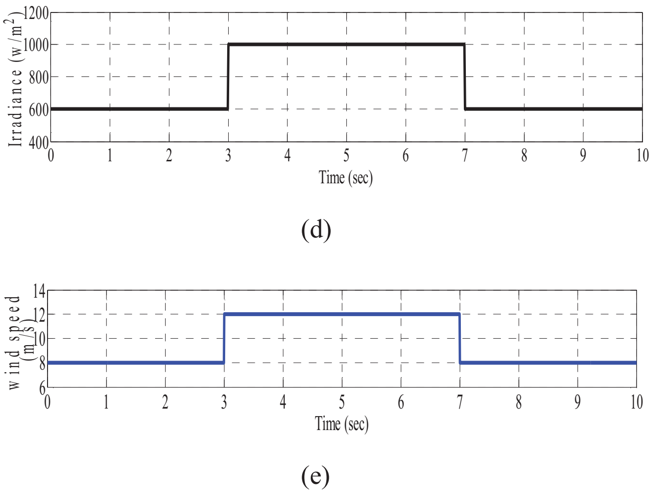


Fig.9: (a) Solar Irradiance (w/m^2), (b) Wind Speed (m/sec)

## VIII. Conclusion

DCMG with various sources under different bus voltage level and CPL is presented in this paper. Constant voltage and maximum power control schemes employed for microsource converters. Autonomous control strategy using decentralized scheme based on the DC bus voltage level is applied for maintaining voltage and power variations under source uncertainties and load deviation. Consequently, the DCMG reliability and power quality is improved. The approximate equivalent transmission line circuit modeled for each subsystem and overall simplified source circuit is analyzed. Based on the overall simplified modeled equivalent circuit, stability analysis of the system is performed using Lyapunov stability criteria to define the qualitative performance of DCMG for satisfactory operation. The controller methodologies are proved with MATLAB/SIMULILK. Therefore this research addresses the effectiveness of decentralized control approach and stability conditions for DCMG system with CPLs.

## IX. APPENDIX

TABLE 2. SYSTEM PARAMETERS USED IN THE SIMULATION

| Subsystem | Parameters | Value | Unit |
|---|---|---|---|
| G-VSC | Power rating | 5 | kW |
| | Switching frequency | 5 | kHz |
| | Filter inductances ($L_1$,$L_2$) | 21.1/6.1 | mH |
| | Damping resistance | 22 | Ω |
| | Filter capacitor | 1 | µF |
| | Grid voltage (rms) | 230 | V |
| Boost converter (Wind ) | Nominal power | 6 | kW |
| | Switching frequency | 5 | kHz |
| | Input inductor | 6.5 | mH |
| | Output capacitor | 140 | µF |
| | Wind input voltage (dc) | 440 | V |
| Boost converter (PV) | Nominal power | 12 | kW |
| | Switching frequency | 2 | kHz |
| | Input inductor | 7.7 | mH |
| | Output capacitor | 470 | µF |
| | PV module output voltage | 320 | V |
| Battery | Nominal voltage | 240 | V |
| | Input inductor | 5.1 | mH |
| | Output capacitor | 200 | µF |
| | Ampere hour (Ah) | 100 | Ah |
| Equivalent Transmissi on line data | Inductance | 50 | µH |
| | Capacitance | 1100 | µF |
| | Resistance | 0.1438 | Ω |
| Load power | | 2.24 ~17 | kW |

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